

Headache, migraine, and structural brain lesions and function: population based Epidemiology of Vascular Ageing-MRI study

SCIFACT-EVIDENCE-17930286

OBJECTIVE To evaluate the association of overall and specific headaches with volume of white matter hyperintensities, brain infarcts, and cognition. **DESIGN** Population based, cross sectional study. **SETTING** Epidemiology of Vascular Ageing study, Nantes, France. **PARTICIPANTS** 780 participants (mean age 69, 58.5% women) with detailed headache assessment. **MAIN OUTCOME MEASURES** Brain scans were evaluated for volume of white matter hyperintensities (by fully automated imaging processing) and for classification of infarcts (by visual reading with a standardised assessment grid). Cognitive function was assessed by a battery of tests including the mini-mental state examination. **RESULTS** 163 (20.9%) participants reported a history of severe headache and 116 had migraine, of whom 17 (14.7%) reported aura symptoms. An association was found between any history of severe headache and increasing volume of white matter hyperintensities. The adjusted odds ratio of being in the highest third for total volume of white matter hyperintensities was 2.0 (95% confidence interval 1.3 to 3.1, P for trend 0.002) for participants with any history of severe headache when compared with participants without severe headache being in the lowest third. The association pattern was similar for all headache types. Migraine with aura was the only headache type strongly associated with volume of deep white matter hyperintensities (highest third odds ratio 12.4, 1.6 to 99.4, P for trend 0.005) and with brain infarcts (3.4, 1.2 to 9.3). The location of infarcts was predominantly outside the cerebellum and brain stem. Evidence was lacking for cognitive impairment for any headache type with or without brain lesions. **CONCLUSIONS** In this population based study, any history of severe headache was associated with an increased volume of white matter hyperintensities. Migraine with aura was the only headache type associated with brain infarcts. Evidence that headache of any type by itself or in combination with brain lesions was associated with cognitive impairment was lacking.